IndoML 2026 Datathon: Technical Report

Noise event detection (Track 1) and noise removal (Track 2) on Vaani speech

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1. Summary

Team sonicboom's Track 1 entry ranks 4th of 82 (1.6112) and has the highest event F1 on the final board (0.7911). Our Track 2 entry ranks 10th of 20 (15.31). For each track we report two systems: a model-only system, which runs one learned pipeline on every clip and reads no file name, peak level or noise library at test time, and our leaderboard entry.

Track	System	What it is	Score	Final rank
1 Event detection	950232, entry	ATST-Frame ensemble + template matching with the noise library	1.6112 (F1 0.7911 + Dice 0.8201)	4th of 82
1 Event detection	C8, model-only	4 ATST-Frame members × 4 time shifts, one vote	1.5925 (F1 0.7795 + Dice 0.8130)	–
2 Noise removal	E3, entry	ZipEnhancer + TF-GridNet + fusion head, with a route rule and a pass-through step	15.31 (SI-SDR 13.20 dB + ΔWER 2.11)	10th of 20
2 Noise removal	T4, model-only	4-view ZipEnhancer + TF-GridNet + fusion head	Validation: 15.62 dB SI-SDR on VH695	–

Track 1 score = event F1 (onset and offset within 20 % of the reference length, at least 50 ms) + segment Dice (10-ms frames); maximum 2.0. Track 2 score = SI-SDR on synthetic clips (dB) + ΔWER, the WER drop with the required SraVaani-1.0 ASR (0.32 × synthetic + 0.68 × natural).

2. Data: how the labels are made

The natural labels follow the transcript, not the sound, so a model must learn the labelling convention from speech context. The training release has 90,630 clips (154.6 h, 59 languages) in three tiers: Gold (11,106 clips, verified times), Silver (61,640 clips, unverified times) and Bronze (17,884 clips, transcript tags only). The validation set has 1,806 natural and 695 synthetic clips.

- **Events come from transcript tags.** In 1,803 of 1,806 validation natural clips, the event list equals the sequence of noise tags in the transcript. A tag wraps words (40 %), marks a point (41 %) or wraps nothing (19 %). Label edges sit at the edges of speech pauses, not at the edges of the noise (spectral flux is at chance level), so edges cannot be snapped to the audio.
- **Long events decide Dice.** Events of 1 s or more are 22.6 % of the validation natural events but 68.5 % of the noise time. 175 of the 376 references of 2 s or more span the whole clip.
- **Silver has a longer style.** It uses the same tags, but its events are longer (median 0.98 s against 0.45 s in Gold), 39 % of its clips have a whole-clip event, and it omits many short events.
- **Synthetic clips reuse a small library.** They are exact sums of clean speech and noise, and their 2,213 validation events use about 203 noise waveforms.
- **Test natural clips are longer** than validation natural clips (mean 7.14 s against 5.00 s). Gold has the test length distribution.

3. Track 1: noise event detection

C8 runs 16 views of each clip (4 members × 4 time shifts) and keeps the events that at least half of the views agree on. 950232 adds template matching with the noise library to a larger ensemble.

Model. Each member is an ATST-Frame encoder, pre-trained on AudioSet, with a RED-style interval head. For each 40-ms frame and each of the 7 classes, the head predicts the presence of an event and the distances to its onset and offset. The four members differ in how they use the weaker tiers:

- **P2** learns from the Gold labels.
- **S1** continues P2 on the Silver times as strong labels (loss weight 0.4).
- **R10** continues P2 on Silver with a target per event: full targets for whole-clip events and events of 2 s or more, presence only for short point tags, and no target for short wraps. Bronze clips whose tag wraps every word add whole-clip events.
- **PSR10B** uses the full Silver objective plus Bronze.

Training. Each member then trains in rounds of 4,000 updates (AdamW; learning rate 5e-6 for the encoder and 5e-5 for the head): three rounds for P2 and S1, two for R10 and PSR10B. Each update mixes 8 Gold clips, 1 clip built

from Gold events, 8 LIBSYN clips (5 in the first round), 8 Silver clips (weight 0.4) and 32 Bronze clips with tags only (weight 0.1). The rounds take 17.5 GPU-hours on Kaggle T4 GPUs.

LIBSYN is our synthetic training audio. A generator places the about 203 library noise waveforms on 25,280 cleaned Vaani speech beds (24.7 h). It follows the placement laws measured on validation: 1 to 5 events per clip, vocal sounds in digital silence, a per-template SNR, no overlaps and gaps of 0.2 s or more. Version 1 has 16,000 clips (28.7 h) and version 2 has 32,000 (57.6 h); each passes nine calibration checks against validation. No validation session is in any training clip or bed, and at test time a model sees only the clip.

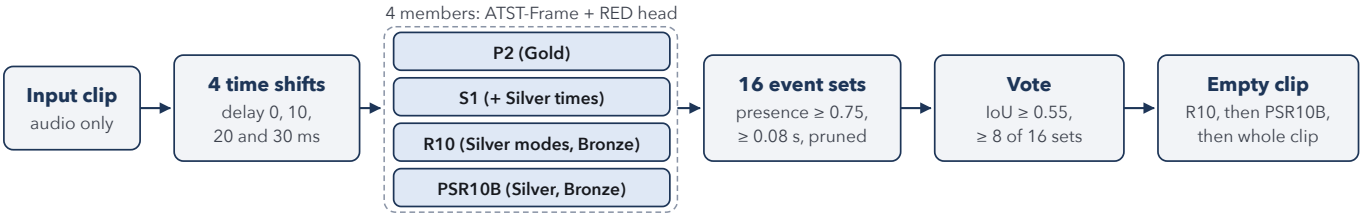


Figure 1. C8 inference. Each member runs on 4 shifted copies of the clip; an event stays when 8 or more of the 16 sets agree.

Inference of C8:

1. Each member runs on 4 copies of the clip, delayed by 0, 10, 20 and 30 ms. This moves the 40-ms frame grid under the signal and gives 16 event sets.
2. In each set, frames with presence 0.75 or more give events of 0.08 s or longer. Weak events (score below 0.70) and overlaps (IoU 0.7) are removed. An empty set is decoded again at 0.65.
3. The vote clusters the events of the 16 sets (IoU 0.55). A cluster stays when 8 or more sets support it. Its onset and offset are the score-weighted means, and the class is not used.
4. An empty clip is decoded again from R10 at 0.55, then from PSR10B. A clip that is still empty gets one event over the whole clip.

Entry 950232 uses a 24-vote ensemble of the same four members, trained on the Vaani tiers only, and a WavLM-Large end refiner. It adds steps that use the organizers' noise library: a router sends each synthetic clip to template matching (normalised cross-correlation) and a whitened matched filter, natural clips get exact library copies, and small rules handle empty clips, synthetic edges and long events.

Server result	C8 (model-only)	950232 (entry)
Score	1.5925	1.6112
Event F1 / Dice	0.7795 / 0.8130	0.7911 / 0.8201
Natural clips: F1 / Dice	0.4924 / 0.7056	0.5192 / 0.7117
Synthetic clips: F1 / Dice	0.9953 / 0.9903	0.9995 / 0.9993

What the parts add (validation). Silver as strong labels (S1) adds +0.011 on the natural clips, and the recall of events of 2 s or more goes from 0.266 to 0.362. One LIBSYN round cuts the synthetic misses from 167 to 15 and the false alarms from 124 to 31. Decoding is not the limit: 27 decode settings on fixed models change the cross-validated score by 0.0000, and 58 to 76 % of the missed references have no support in any member. Note: 87 library templates also occur in 77 test natural clips, so LIBSYN can teach those events; this is worth up to about +0.004.

4. Track 2: noise removal

T4 fuses two fine-tuned enhancers with a learned time-frequency head and averages the main enhancer over four views of the clip. E3 adds a route rule and a pass-through step.

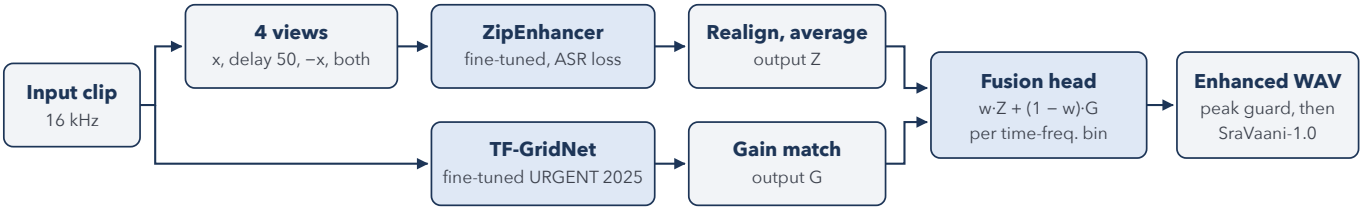


Figure 2. T4 inference. The ZipEnhancer branch is averaged over 4 views; a learned head mixes it with TF-GridNet in each time-frequency bin.

Pipeline of T4:

1. ZipEnhancer (2.0 M parameters) cleans 4 views of the clip: the clip, the clip delayed by 50 samples, the sign-flipped clip and both. The outputs are realigned, sign-restored and averaged.
2. TF-GridNet (8.5 M parameters) cleans the clip. A least-squares gain matches its level to the ZipEnhancer output.

3. A fusion head (29k parameters) predicts a weight w for each time-frequency bin: output $= w \times$ ZipEnhancer $+ (1 - w) \times$ TF-GridNet. The last 64 samples use ZipEnhancer only.
4. The clip is scaled down if it would clip int16. Clips over 20 s run in 20-s windows with 1-s cross-fades.
5. SraVaani-1.0 transcribes the enhanced audio.

Training targets. The organizer clean references are gated speech (median 10 % exact-zero samples), but raw Vaani speech keeps its background, and fine-tunes on raw speech lose 1.2 to 2.2 dB. So our targets are clean beds: DeepFilterNet3 cleans event-free Vaani speech spans (Gold and Silver), and a Silero VAD gate zeroes the pauses. This gives 25,280 beds (24.7 h) with no validation session.

Noise and mixtures. Public sets give 40 % of the noise (DEMAND, ESC-50 folds 1 to 4, MUSAN noise and music without vocals), Vaani point-tag sounds 25 %, procedural noise 20 % (engines, fans, hiss, tones) and Vaani environmental events 15 %. Only noise without speech is kept (VAD filter), because noise with words teaches the model to remove the target speech (−2.5 dB). Each 2-s crop gets 1 to 3 events at an SNR drawn from $N(-8, 6)$ dB, clipped to $[-20, 12]$ dB.

Training. ZipEnhancer starts from the public ModelScope model: 2,000 steps with an SI-SNR loss, then 3,000 steps with SI-SNR plus L1 losses on the frozen SraVaani front-end features and encoder outputs (weight 0.25 each). TF-GridNet starts from the ESPnet URGENT 2025 model and is fine-tuned with SI-SNR on 2-s crops. The fusion head trains for 20 epochs (negative SI-SNR) on the outputs of both enhancers for 6,000 new mixtures. All training runs on an Apple M-series GPU; a softplus form of the ZipEnhancer SwooshL/R activation removes non-finite gradients.

Entry E3 uses the same models with one ZipEnhancer view, a route rule (ZipEnhancer alone on mostly clean clips) and a pass-through step (the input stays in 16-ms frames that the model changes by less than 20 dB). It scores 15.31 on the server: SI-SDR 13.20 dB + ΔWER 2.11 (+6.28 synthetic, +0.14 natural).

Validation set	Noisy input	T4 (model-only)
VH695: the 695 synthetic pairs, noise × 1.68 (test level)	SI-SDR −4.84 dB, WER 36.75 %	SI-SDR 15.62 dB, WER 32.64 %
VS695: the same pairs at the organizers' level	SI-SDR −0.33 dB	SI-SDR 18.78 dB
NAT1806: the 1,806 natural clips	WER 28.16 %	WER 28.28 %

What the parts add (validation). On VH695, the ASR-feature stage lowers the WER of the ZipEnhancer branch from 34.41 % to 33.09 % and raises its SI-SDR by 0.44 dB. The fusion head adds 0.145 dB over a fixed 0.7/0.3 average. The four-view average adds 0.25 dB on VH695 (+0.137 dB, $t = 9.2$, on the 558 clips with the most noise) and lowers its WER by 0.29 points.

5. Code, pre-trained models and data

The code for both tracks is at github.com/Ayush-004/indoml-2026-datathon. Each track folder has a README with the setup, training and inference steps and the full list of dependencies.

Dependency	Source	Use
ATST-Frame encoder (AudioSet)	PretrainedSED	Track 1 members
WavLM-Large	Microsoft, on Hugging Face	Track 1 end refiner of 950232
ZipEnhancer	ModelScope	Track 2 main enhancer
TF-GridNet, URGENT 2025	ESPnet; weights by Kohei Saijo	Track 2 second enhancer
SraVaani-1.0 ASR	ARTPARK-IISc, on Hugging Face	Track 2 transcripts and the ASR-feature loss
DeepFilterNet3, Silero VAD	Public releases	Track 2 clean speech beds
Vaani Noise-Event Dataset; DEMAND, ESC-50, MUSAN	Hugging Face; public noise sets	Training data

Track 1 members train on Kaggle T4 GPUs. Track 2 models train and run on an Apple M-series laptop. Our fine-tuned weights are in the v1.0 release of the repository.